

DR3500CA Rear Helper Spring Kit Installation Manual

Model: DR3500CA (Rear Suspension Package)

Please review these instructions thoroughly prior to commencing installation. Ensure all standard safety protocols are observed when performing work beneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Rubber Spacer (4" O.D. x 1" Thick with AH Bushing)	2
3	Hollow Rubber Spring	2
4	Hex Cap Screw (HCS) (3/8" x 1-3/4" UNC GR 5 Z)	2
5	Hex Cap Screw (HCS) (3/8" x 2-3/4" UNC GR 5 Z)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the axle assembly to hang freely. Securely support the chassis before proceeding with any work.
- 2. Remove OEM Bump Stop:** Detach and remove the vehicle's original equipment manufacturer (OEM) bump stop.
- 3. Assemble Spring and Bracket:** Integrate the hollow rubber spring with its corresponding mounting bracket. Refer to any provided diagrams for correct orientation.
- 4. Mount Assembly:** Secure the assembled helper spring unit to the vehicle frame utilizing the existing OEM fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to its resting position on the ground.

Installation Notes:

- A rubber spacer (Ref. 2) and a longer hex cap screw (Ref. 5) may be utilized to achieve a specific desired ride height.
- When disassembling OEM fasteners, the application of penetrating fluid is strongly recommended to facilitate removal.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** For light-duty trucks and vans, an optimal clearance of 1/2 inch to 1-1/2 inches should be maintained between the helper spring and its contact surface when the vehicle is unladen.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will diminish or entirely close this initial clearance gap.
- **Heavy-Duty Applications:** For medium-duty and heavy-duty trucks, trailers, and motorhomes, the helper spring assembly is designed to maintain constant contact with no discernible gap.

Important Safety Warnings:

- Do not exceed the maximum axle weight rating specified by the vehicle manufacturer.

- Inspect for potential interference between the helper spring and surrounding vehicle components. Note that under load, the rubber spring element will experience an increase in its effective diameter.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.